

KWAME NKRUMAH UNIVERSITY OF SCIENCE & TECHNOLOGY, KUMASI
DEPARTMENT OF COMPUTYER SCIENCE
SECOND SEMESTER, 2025/2026 ACADEMIC YEAR

COURSE OUTLINE: MATH 162				
Course Title	Introduction to Pure Mathematics II			
Course Code	MATH 162			
Contact Hours	3 hours consisting of 2 hours of lecture and an hour of tutorial.			
Course Instructor	Kwame Piesie Frimpong, PhD			
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Phone/WhatsApp	0243 33 76 03			
Office	Casely-Hayford Building, 4th Floor Room 434			
Office Hours	Fridays from 10 hours GMT to 14 hours GMT			
Examination	Mid-semester exam and End of semester exam will be done online.			
Course objectives	The course is intended to offer students the basic skills in Calculus and Geometry. It is designed to provide students who intend to do further Mathematics or a course in Mathematical Science a strong foundation in analysis. It is also aimed at introducing the Mathematical concepts behind the rules and methods studied in Calculus. This course is a prerequisite for students who may pursue graduate degrees in Mathematics, Statistics and allied Science programmes.			
Learning Outcomes	By the end of the semester, students should be able to - understand the notion of a function, limit of functions and the idea of continuity - understand L'Hospital's rule/concept of limit of a rational function and indeterminate forms, - grasp the idea of derivative of a function, appreciate various rules of differentiation and differentiation from first principles, - have a feel of how to differentiate certain function: polynomial, rational, trigonometric, hyperbolic, exponential, logarithm etc. - get an idea of the fundamental definition of differentiability at a point, - have a feel of various applications of differentiation, - come to terms with certain fundamental theorems of calculus, - appreciate the meaning and application of integration, - familiarize themselves with coordinate and analytic geometry as a discipline in Mathematics; specifically, understand the details of the following curves: circle, parabola, hyperbola, ellipse etc.			
Assessment Methods	Overall Assessment (100%)	Assignments (10%)	Mid-Semester Exam (30%)	End of Semester Exam (60%)
Mode of Delivery	1. Lectures	2. Tutorials	3. Trial Exercises	4. Assignments
Topics & Weeks	Lecture Course		Venue	Assessment
1. The concept of a function	Definition of a function, Domain and Range of a function, Types of functions (Even, Odd, Polynomial, Rational, Periodic etc)		SF 1	Test of knowledge Exercise
2. Continuation of functions	- A look at these functions: Trigonometric, Exponential, Hyperbolic, Logarithm etc), - Identities existing between Trig, Exponential and Hyperbolic Functions - Osborn's rule.		SF 1	Test of knowledge Exercise

3. Limit of functions and Continuity	- The idea of limit of a function, Computation of limit of a function. - L'Hospital's concept of limit computation, The notion of continuity.	SF 1	Assignment 1
4. The derivative of a function	- Basic definition of differentiation using limit. - Rules of differentiation and differentiation of some selected types of functions.	SF 1	Test of knowledge Exercise
5. Higher-order derivatives	A general look at higher-order derivatives	SF 1	Trial Exx
6. Implicit differentiation	Polynomials, Trigonometric etc	SF 1	Assignment II
7. Revision	Revision and Mid-Semester Examination	SF 1	Test of knowledge Exercise
8. Parametric differentiation	Parametric equations in general: Polynomials, Trigonometric, Exponential and Hyperbolic.	SF 1	Mid Sem
9. Derivative of inverse functions	Trigonometric functions, Hyperbolic functions	SF 1	Test of knowledge Exercise
10. Integration	Methods of integration (polynomials, trigonometric, hyperbolic, exponential, natural logarithm etc.)	SF 1	Assignment III
11. Applications of differentiation	- The gradient function and the gradient of tangents and normals at a point. - Equations of tangents and normals	SF 1	Test of knowledge Exercise
12. Applications of integration	Rates of change, Kinematics and Small changes etc.	SF 1	Assignment IV

References:

- Ron, L. and Bruce H. E. (2009), Calculus, Publisher: Cengage Learning.
- Stewart, J. (2011), "Calculus". (International metric edition): ISBN-13: 978-053849884, Brook/Cole Publishing).
- Michael Spivak (2006) Calculus 3rd Edition. ISBN-13: 978-0521867443, Cambridge University Press.
- You may download from www.google.com any good material and any **YouTube** video on Calculus.

Students Responsibility

- Students should be punctual and participate actively in class discussions.
- Prompt and timely delivery of assignments.
- It is the policy of the University that students fulfill the attendance requirement of at least 80%. Any student who contravenes this requirement will not be allowed to write the final exam. Such a student will be graded F.**

Counsel

- Students are hereby advised that success is neither a fortuitous combination of circumstances nor a serendipitous discovery. It is not achieved on a silver platter. It is indeed a result of **Grace** and proper application of well nurtured potential.

Name: Kwame Piesie**Sign:** *KPF***Date:** May 25, 2026